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To cite this article: Nvzhi Tang *et al* 2020 *J. Phys.: Conf. Ser.* **1617** 012036

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Research on Development and Application of Underwater Acoustic Communication System

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Abstract—Underwater acoustic communication technology is the core technology in the marine field, occupying a very important position, and has an important supporting role for underwater observation operations and military activities. Underwater acoustic technology is currently the most widely used and longest underwater wireless communication technology. With the increasing demand for development, the complexity of underwater acoustic channels, underwater environment and other factors make the communication speed, communication distance and communication stability of underwater acoustic communication greatly restricted. Despite these limitations, underwater acoustic communication is still valued by professionals from various countries and industries, and has been developed and applied to some extent. This article summarizes the basic content of underwater acoustic communication technology, leads to its general structure as a communication system, the current development of underwater acoustic communication technology, and the characteristics of underwater acoustic communication systems due to the characteristics of underwater acoustic channels. This article summarizes the development and some applications of the underwater acoustic communication system and the full text and some prospects for the future underwater acoustic communication.

1. INTRODUCTION

2020 is the era of information and network development. Communication technology is an important guarantee to support the rapid development of e-commerce, smart home, telemedicine, cloud computing, cloud storage and other industries. The wireless communication technology represented by 5G can reach up to 10Gbps, and its low-cost advantage will make the communication network cover the world. Because the sound wave can travel a long distance in the water, underwater acoustic communication is an underwater wireless information transmission technology. However, it is also faced with such factors as narrow available bandwidth, complex and changeable channel, and faces great challenges [1]. In spite of this, various countries have made great efforts to study underwater acoustic communication technology, which has great contribution value for national military and ocean exploration. The technical personnel and institutions who study and engage in underwater acoustic communication technology also make unremitting efforts to overcome the difficulties, which makes the underwater acoustic communication technology has a good development, and has a significant application in environmental survey and other aspects.

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Figure 1. Diagram of reaction flow and product structure

2. UNDERWATER ACOUSTIC COMMUNICATION SYSTEM

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2.1 System model

The working principle of underwater acoustic communication system is to use transducer to convert electrical signal into acoustic signal at transmitting end and convert acoustic signal into electric signal at receiving end. The block diagram of underwater acoustic communication system is shown in Fig. 2.

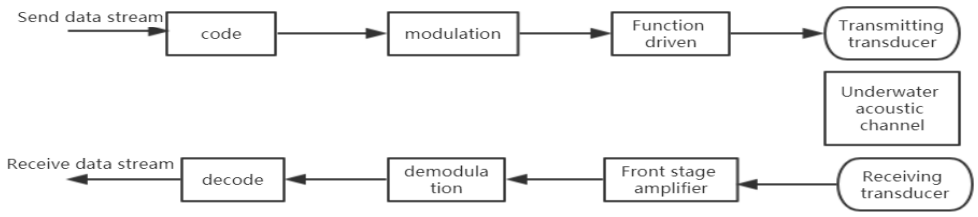


Figure 2: underwater acoustic communication system

2.2 Underwater acoustic channel model

The channel of underwater acoustic signal transmission includes water body, sea surface and seabed, and the characteristics of underwater acoustic communication technology are mainly determined by the characteristics of underwater acoustic channel. Underwater acoustic channel is a kind of parametric channel, and its characteristic parameters change randomly with the change of space, time and frequency. The underwater acoustic channel model is shown in Fig. 3.

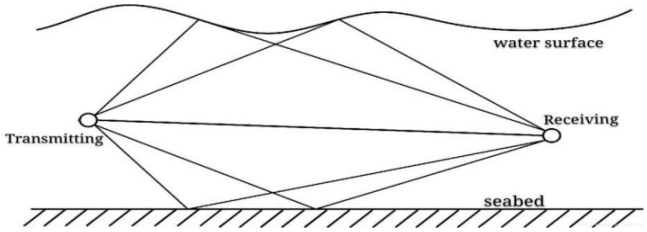


Figure 3 : underwater acoustic channel model

2.3 Characteristics of underwater acoustic channel

Fluctuation effect. The acoustic signal will fluctuate in the transmission process due to the inhomogeneity in space and the randomness in time. The moving sea surface, uneven sea floor and uneven water body are the factors affecting the effect.

Time varying effect. Due to the internal waves, water masses, turbulence and the change of relative position of communication targets, the underwater acoustic channel often shows time-varying characteristics.

Multipath effect. The underwater acoustic signal will be transmitted in different paths after transmitting at the transmitter, and the receiver will receive multiple signals from the same signal through different paths in sequence at different times. The time delay of multipath effect also varies with the depth of water.

Environmental noise. There are many noise sources in the ocean, including the natural noise caused by sea waves and organisms, and the artificial noise caused by ships and industries. These different noises have different noise levels and frequencies, which have different effects on the underwater acoustic signals.

Doppler effect. The relative motion between the receiver and the transmitter changes the frequency of the received signal. The Doppler frequency shift of underwater acoustic signal will be caused by the reflection of sea surface and sea bottom factors and the refraction of turbulence. Therefore, the Doppler frequency shift at the receiving end is non unitary and causes Doppler frequency shift diffusion.

The channel bandwidth is small. Because the absorption and attenuation of sea water to the acoustic signal increases with the frequency index, the bandwidth of transducer is limited, which makes the underwater acoustic signal only use low-frequency signal, the channel bandwidth is small, so the communication rate is relatively low.

3. CURRENT RESEARCH STATUS AND DEVELOPMENT TREND OF UNDERWATER ACOUSTIC COMMUNICATION

3.1 Research status

In recent years, with the support of the "863" plan, the military, and the National Natural Science Foundation, progress has been made in the field of underwater acoustic communication in my country in terms of communication algorithms, communication machine development, network protocol simulation, networking application tests, and protocol specification formulation.

3.2 Future technology development trends

Development of short-range high-speed underwater acoustic communication nodes and networking technology: research and development of short-range high-speed underwater acoustic communication nodes with an operating distance of 1 to 2 km, study of short-range high-speed underwater acoustic communication technologies, and focus on improving performance in shallow sea channels. In order to meet the interaction needs of large amounts of data in shallow sea network observations, study and optimize network protocols and technical solutions [2].

Development of long-range and ultra-long-range low-speed communication technologies. [2-3] Develop long-range and ultra-long-range low-speed communication technologies, increase communication distance and speed, and study network protocols that are compatible with long-range low-speed.

The development of collaborative observation technology that matches the capabilities of the underwater acoustic communication network. [2] Due to the limited channel capacity of the underwater acoustic communication network, the amount of information that can be exchanged between the nodes of the network is small. Therefore, it is necessary to develop collaborative observation technology to improve the data pre-processing capability of the observation station and give full play to the collaborative advantages of network observation.

Maritime demonstration verification test. [2] Combined with the application requirements of ocean observation and so on, offshore demonstration and verification tests were carried out for different types of experimental sea areas such as offshore and deep sea. The underwater acoustic communication network and the submarine observation network are combined to achieve better connection and

optimization between the underwater acoustic communication network and the submarine optical cable network, and to increase the flexibility of the deep-sea observation network.

4. APPLICATION OF UNDERWATER ACOUSTIC COMMUNICATION

4.1 Multipoint monitoring system of marine water quality based on underwater acoustic communication

In the process of marine aquaculture, the past water quality monitoring has a long monitoring period, high cost, a lot of manpower and material resources, and it cannot monitor the dynamic changes of the water environment in real time. The researchers designed a multi-point detection system for marine water quality with the aim of monitoring the seabed and sea level water quality [5-9]. This system uses marine buoys to mount water quality sensors to measure sea level water quality parameters and buoys to mount DTUs. The dual communication with Beidou satellite realizes the data uploading in the near sea. It can be seen that the water quality monitoring system designed based on hydroacoustic communication technology is feasible and has important significance for marine aquaculture.

4.2 Environmental monitoring of seabed natural gas hydrate based on hydroacoustic communication

Submarine gas hydrate environmental monitoring mainly collects, transmits, processes, and analyzes submarine data and information. The principle of the submarine gas hydrate environmental monitoring system [10-14] is to use underwater acoustic modem to achieve system communication. The underwater acoustic modem transmits the data collected by the monitoring and control system placed on the seabed to the water surface base station, and the water surface base station then transmits the data through satellite communication. Forward to the land control center. It can be seen that the underwater acoustic communication system has obtained very good research and application value in the environmental monitoring of submarine gas hydrates.

4.3 Solving the problem of data transmission while drilling using the principle of underwater acoustic communication

The researchers proposed the idea of using underwater acoustic communication principles to solve the problem of data transmission while drilling. The drilling fluid in the drill pipe is selected as the underwater acoustic communication channel, and it is simplified into a one-dimensional, finite-length, uniform, and equal cross-section waveguide. According to the small amplitude wave equation of the ideal fluid, the influence of factors such as the cut-off frequency, viscous force and nonlinear characteristics of the acoustic channel while drilling is studied, and the basic parameters such as frequency, power and transmission distance of the communication system are finally determined. The researchers simulated and analyzed the transmission characteristics of the water-while-drilling acoustic channel [15-19], and finally quantitatively calculated the attenuation of the water-based drilling fluid and oil-based drilling fluid to the underwater acoustic signal, and proved through a series of experiments that the principle of using underwater acoustic communication technology. It is feasible to solve the problem of high-speed data transmission while drilling, which can actually solve the problem of data transmission while drilling.

5. CONCLUSIONS

In recent years, with the continuous innovation and development of communication technology, it has been well developed in many fields, and its communication quality and data transmission speed have also been significantly improved. Although the current development of underwater acoustic communication is affected by the seawater environment and the complexity of underwater acoustic channels, underwater acoustic communication is of great significance to the marine survey. Various countries attach great importance to the development of underwater acoustic communication technology and continuously overcome the difficulties of underwater communication problem. Therefore, underwater acoustic communication has been well developed and applied in various

aspects, such as the three applications illustrated in the article. In the future, underwater acoustic communication will continue to become a key development technology in various countries. Under the guidance of 5G technology, some of the limiting factors of underwater acoustic communication will continue to be reduced. Underwater acoustic communication technology will usher in better development and application in the future, and will gain greater development space in the construction of future communication projects.

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